

Source-labelled development archive

2,160 variants across 17 genes
1,187 B/LB and 973 LP/P

**Training-only feature construction**

47 semantic candidates; Reliability_index excluded
Missingness screen, imputation and scaling within training

**Repeated nested cross-validation**

5 repeats × 5 outer folds; 4 inner folds
Feature selection refitted inside each inner-training fold
SVM-RBF · L2 logistic · Extra Trees · HistGB

**Frozen deployment model**

25 predictors; four learners, calibrators and thresholds
S_MAC parameters and fitted objects retained

**Identical audit output for every evaluable record**

Four probabilities and votes · four-bit pattern · V0-V4 · S_MAC
V0: unanimous B/LB-directed | V1-V3: mixed votes | V4: unanimous LP/P-directed
Below 80% observed panel features: Incomplete; predictions suppressed

Gene-held-out audit

17 whole-gene holdouts
Per-gene estimates

**Sensitivity analyses**

Reliability_index inclusion
addAF + AlphaMissense
exclusion; transcript scope

**Application catalogue**

43,679 unresolved records
No outcome labels

